

MODULE – IV COMPLETE ANSWERS (DBMS)

Based on the uploaded question bank.

SHORT ANSWER QUESTIONS (5 MARKS)

1. Explain Functional Dependency (FD) with Suitable Example

A Functional Dependency (FD) is a relationship between attributes in a relation where one attribute uniquely determines another attribute.

Definition

If attribute **A** determines attribute **B**, then:

$A \rightarrow B$

Read as:

A determines B

Example

STUDENT

SID	Name	Dept
101	Anik	CSE
102	Rahul	EE

Since SID uniquely identifies a student,

$SID \rightarrow Name, Dept$

Meaning:

Knowing SID allows us to determine Name and Department.

Types

1. Trivial FD
 2. Non-Trivial FD
 3. Fully Functional Dependency
 4. Partial Dependency
 5. Transitive Dependency
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2. Explain Transitive Dependency with Example

A transitive dependency exists when:

- A determines B
- B determines C

Then A indirectly determines C.

Representation

$A \rightarrow B, B \rightarrow C \Rightarrow A \rightarrow C$

Example

EMPLOYEE

EmpID	DeptID	DeptName
101	10	CSE

FDs:

$EmpID \rightarrow DeptID$

$DeptID \rightarrow DeptName$

Therefore:

$\text{EmpID} \rightarrow \text{DeptName}$

This is a transitive dependency.

Problem

Causes redundancy.

Solution

Normalize to 3NF.

3. Define Third Normal Form (3NF) with Example

A relation is in 3NF if:

1. It is already in 2NF.
 2. No transitive dependency exists.
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Example

EMPLOYEE

EmpID	EmpName	DeptID	DeptName
e			

FDs:

$\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$

$\text{DeptID} \rightarrow \text{DeptName}$

Transitive dependency exists:

$\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}$

Convert into

EMPLOYEE

EmpID	EmpName	DeptID
	e	

DEPARTMENT

DeptID	DeptName
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Now transitive dependency is removed.

Hence relation is in 3NF.

4. What is Redundancy? Explain Anomalies

Redundancy

Redundancy means unnecessary duplication of data.

Example

SID	Name	DeptName
101	Anik	CSE
102	Rahul	CSE
103	Priya	CSE

"CSE" is repeated many times.

Anomalies Caused by Redundancy

1. Insertion Anomaly

Cannot insert department without student information.

2. Update Anomaly

Changing department name requires multiple updates.

Example:

CSE → Computer Science

Must update every record.

3. Deletion Anomaly

Deleting last student of a department may remove department information.

Solution

Normalization.

5. Classification of Functional Dependency

1. Trivial FD

Right side attribute already present in left side.

Example:

$AB \rightarrow A$

2. Non-Trivial FD

Right side attribute absent from left side.

Example:

$A \rightarrow B$

3. Completely Non-Trivial FD

No common attributes.

Example:

$A \rightarrow BC$

4. Full Functional Dependency

Entire key needed.

Example:

$(\text{StudentID}, \text{CourseID}) \rightarrow \text{Marks}$

5. Partial Dependency

Only part of composite key determines attribute.

Example:

$\text{StudentID} \rightarrow \text{StudentName}$

6. Transitive Dependency

Example:

$\text{EmpID} \rightarrow \text{DeptID}$

$\text{DeptID} \rightarrow \text{DeptName}$

6. Candidate Keys of R(A,B,C,D,E,F)

Given FDs:

$A \rightarrow B$

$C \rightarrow DF$

$AC \rightarrow E$

$D \rightarrow F$

Step 1: Find Closure of AC

Start:

$(AC)^+ = \{A, C\}$

Using:

$A \rightarrow B$

Add B

$\{A, B, C\}$

Using:

$C \rightarrow DF$

Add D, F

$\{A, B, C, D, F\}$

Using:

$AC \rightarrow E$

Add E

$\{A, B, C, D, E, F\}$

All attributes obtained.

Hence:

$(AC)^+ = ABCDEF$

Candidate Key

AC

7. Define Fourth Normal Form (4NF)

A relation is in 4NF if:

1. It is in BCNF.
 2. No non-trivial multivalued dependency exists.
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Example

STUDENT

Student	Hobby	Language
A	Music	English
A	Music	Hindi
A	Dance	English
A	Dance	Hindi

Multivalued dependencies:

Student $\twoheadrightarrow$ Hobby

Student $\twoheadrightarrow$ Language

Decompose Into

STUDENT_HOBBY

| Student | Hobby |

STUDENT_LANGUAGE

| Student | Language |

Now relation is in 4NF.

8. Explain Join Dependency and Fifth Normal Form (5NF)

Join Dependency

A relation can be reconstructed by joining multiple projections.

Example

SPJ

| Supplier | Part | Project |

Can be decomposed into:

SP

SJ

PJ

And reconstructed through joins.

Fifth Normal Form (5NF)

A relation is in 5NF if:

- Every join dependency is implied by candidate keys.

Advantages

- Eliminates redundancy.
 - Eliminates join anomalies.
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9. Find Closure of $\{XY\}$

Given:

$z \rightarrow y$
 $y \rightarrow z$
 $x \rightarrow y$
 $x \rightarrow v$
 $vw \rightarrow x$

Find:

$(XY)^+$

Start

$\{X, Y\}$

Using:

$X \rightarrow Y$

Already present.

Using:

$Y \rightarrow Z$

Add Z.

Using:

$X \rightarrow V$

Add V.

Final Closure

$(XY)^+ = \{X, Y, Z, V\}$

10. Check Equivalence of FD Sets

Given:

$F = \{$
 $A \rightarrow C,$
 $AC \rightarrow D,$
 $E \rightarrow AD,$
 $E \rightarrow H$
 $\}$

and

$G = \{$
 $A \rightarrow CD,$
 $E \rightarrow AH$
 $\}$

From G

$A \rightarrow CD$

gives:

$A \rightarrow C$
 $A \rightarrow D$

Also:

$E \rightarrow AH$

gives:

$E \rightarrow A$
 $E \rightarrow H$

and

$A \rightarrow D$

Hence:

$E \rightarrow D$

Thus all FDs of F can be derived.

Similarly F derives G.

Therefore

$F \equiv G$

Both FD sets are equivalent.

11. "All BCNF Relations are in 3NF but not Vice Versa"

BCNF Definition

For every FD:

$X \rightarrow Y$

X must be a super key.

3NF Definition

For every FD:

$X \rightarrow Y$

Either:

- X is super key
 - OR
 - Y is prime attribute.
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Example (3NF but not BCNF)

Relation:

$R(\text{Student}, \text{Course}, \text{Instructor})$

FDs:

$(\text{Student}, \text{Course}) \rightarrow \text{Instructor}$

$\text{Instructor} \rightarrow \text{Course}$

Candidate Key:

$(\text{Student}, \text{Course})$

Since Course is prime attribute,

Relation satisfies 3NF.

But:

$\text{Instructor} \rightarrow \text{Course}$

Instructor is not super key.

Therefore:

Not BCNF.

Conclusion

$\text{BCNF} \subset 3\text{NF}$

All BCNF relations are in 3NF, but not all 3NF relations are in BCNF.

12. Explain Multivalued Dependency (MVD)

A multivalued dependency occurs when one attribute determines multiple independent values.

Notation

$A \twoheadrightarrow B$

Example

STUDENT

Student	Hobby	Language
A	Music	English
A	Music	Hindi
A	Dance	English
A	Dance	Hindi

Student independently determines:

Hobby
Language

Therefore:

Student $\twoheadrightarrow$ Hobby
Student $\twoheadrightarrow$ Language

LONG ANSWER QUESTIONS (10 MARKS)

1. Check Lossless Join and Dependency Preservation

Given:

$R(A,B,C,D)$

FDs:

$A \rightarrow B$

$C \rightarrow D$

Decomposition:

$R1(A,B)$

$R2(C,D)$

Lossless Join Test

Intersection:

$R1 \cap R2 = \emptyset$

No common attribute.

Therefore:

Lossy Join

Dependency Preservation

$R1$ preserves:

$A \rightarrow B$

$R2$ preserves:

$C \rightarrow D$

Hence:

Dependency Preserving = YES

2. Find Closure of BG and AF

Given:

$AB \rightarrow CD$

$AF \rightarrow D$

$DE \rightarrow F$

$C \rightarrow G$

$F \rightarrow E$

$G \rightarrow A$

(BG)+

Start:

$\{B, G\}$

Using:

$G \rightarrow A$

Add A.

Now:

$AB \rightarrow CD$

Add C,D.

Using:

$C \rightarrow G$

Already present.

Using:

$DE \rightarrow F$

Need E.

Using:

$F \rightarrow E$

Need F.

Cannot proceed.

Final:

$(BG)^+ = \{A, B, C, D, G\}$

$(AF)^+$

Start:

$\{A, F\}$

Using:

$AF \rightarrow D$

Add D.

Using:

$F \rightarrow E$

Add E.

Using:

$DE \rightarrow F$

Already available.

Final:

$(AF)^+ = \{A, D, E, F\}$

Fully Functional Dependency

A dependency where the whole key is required.

Example:

$(\text{StudentID}, \text{CourseID}) \rightarrow \text{Marks}$

Neither StudentID nor CourseID alone determines Marks.

3. Shipping Relation (Candidate Key & Normalization)

Relation:

Shipping(
ShipName,
ShipType,
VoyageID,
Cargo,
Port,
Date
)

FDs:

$\text{ShipName} \rightarrow \text{ShipType}$

$\text{VoyageID} \rightarrow$
 $\text{ShipName}, \text{Cargo}$

$\text{ShipName}, \text{Date} \rightarrow$
 $\text{VoyageID}, \text{Port}$

Candidate Key

Compute closure:

$(\text{ShipName}, \text{Date})^+$

Gives all attributes.

Hence Candidate Key:

$\{\text{ShipName}, \text{Date}\}$

Normalization

Table 1

SHIP

ShipName
ShipType

Table 2

VOYAGE

VoyageID
ShipName
Cargo

Table 3

ARRIVAL

ShipName
Date
VoyageID
Port

These relations satisfy BCNF.

4. Define BCNF and Difference from 3NF

BCNF

For every FD:

$X \rightarrow Y$

X must be super key.

Difference

3NF	BCNF
Less strict	More strict
Allows some redundancy	Eliminates more redundancy
Easier decomposition	Stronger normalization

Why BCNF is Stronger?

Because every determinant must be a super key.

Hence fewer anomalies.

5. Armstrong's Axioms

Used to infer functional dependencies.

Reflexivity

If:

$$Y \subseteq X$$

Then:

$$X \rightarrow Y$$

Example:

$$AB \rightarrow A$$

Augmentation

If:

$$X \rightarrow Y$$

Then:

$$XZ \rightarrow YZ$$

Example:

$$A \rightarrow B$$

$$AC \rightarrow BC$$

Transitivity

If:

$$A \rightarrow B$$

$$B \rightarrow C$$

Then:

$$A \rightarrow C$$

Derived Rules

Union

$$A \rightarrow B$$

$$A \rightarrow C$$

$$\Rightarrow A \rightarrow BC$$

Decomposition

$$A \rightarrow BC$$

$\Rightarrow A \rightarrow B$
 $A \rightarrow C$

Pseudotransitivity

$A \rightarrow B$
 $BC \rightarrow D$

$\Rightarrow AC \rightarrow D$

6. SQL Queries

Relations:

Works(Pname,Cname,Salary)

Lives(Pname,Street,City)

Located_In(Cname,City)

Manager(Pname,MgrName)

i) Persons living in Bangalore

```
SELECT Pname
FROM Lives
WHERE City='Bangalore';
```

ii) Infosys employees with salary between 50000 and 100000

```
SELECT Pname
FROM Works
WHERE Cname='Infosys'
AND Salary BETWEEN 50000 AND 100000;
```

iii) Persons who live and work in same city

```
SELECT W.Pname
FROM Works W, Lives L, Located_In LI
WHERE W.Pname=L.Pname
AND W.Cname=LI.Cname
AND L.City=LI.City;
```

iv) People working for Tech M and their cities

```
SELECT W.Pname,L.City
FROM Works W, Lives L
WHERE W.Pname=L.Pname
AND W.Cname='Tech M';
```

v) Average salary of Infosys employees

```
SELECT AVG(Salary)
FROM Works
WHERE Cname='Infosys';
```

MODULE IV EXAM REVISION

Most Important Topics:

1. Functional Dependency (FD)
2. Closure of Attributes
3. Candidate Key
4. 1NF, 2NF, 3NF
5. BCNF
6. Armstrong's Axioms
7. Multivalued Dependency
8. 4NF and 5NF
9. Lossless Join
10. Dependency Preservation

Very Important for University Exams:

- Candidate Key Numericals
- Closure Numericals
- BCNF vs 3NF

- Armstrong's Axioms
- Normalization Steps

These questions are among the highest-scoring and most frequently repeated topics in DBMS examinations.